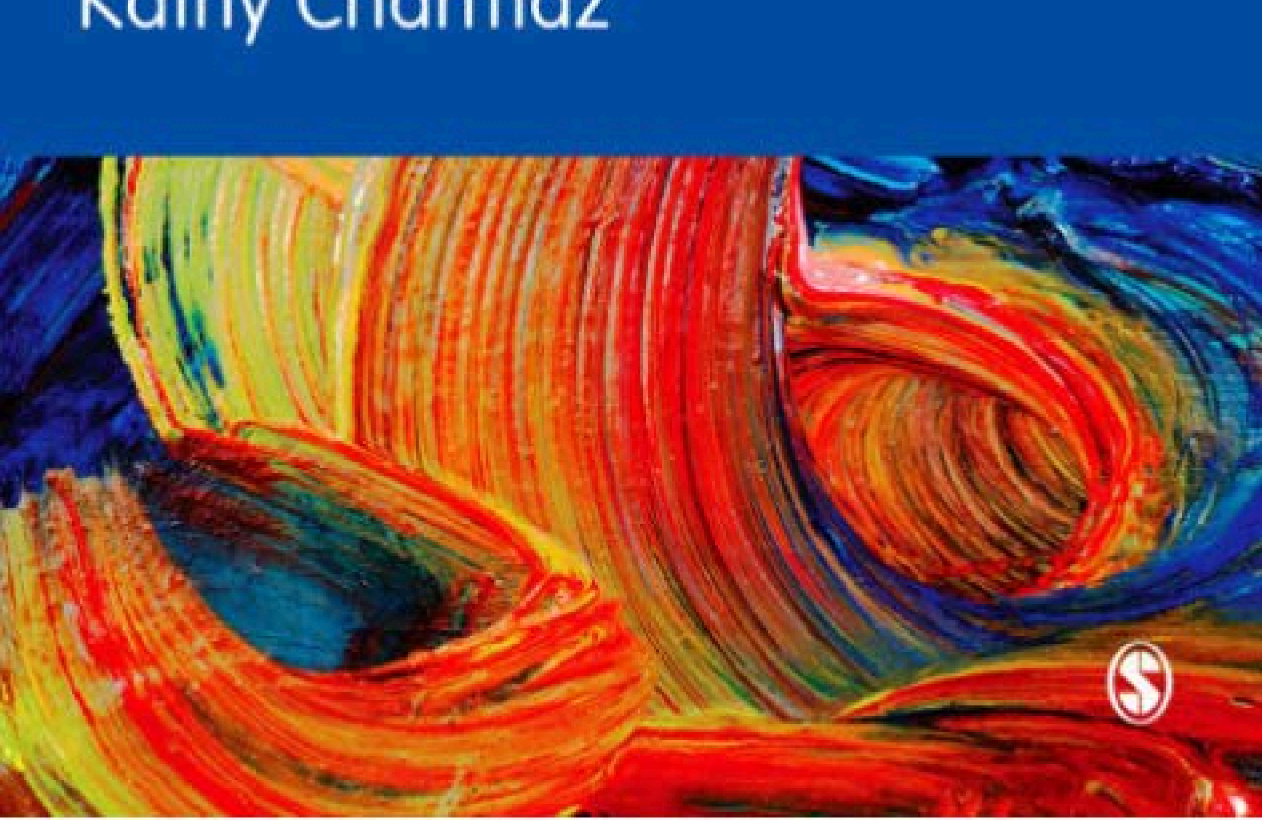
Constructing Grounded Theory

2nd Edition

Kathy Charmaz



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The Logic of Grounded Theory Coding Practices and Initial Coding

The first analytic turn in our grounded theory journey brings us to coding. Grounded theory coding requires us to stop and ask analytic questions of the data we have gathered. These questions not only further our understanding of studied life but also help us direct subsequent data-gathering toward the analytic issues we are defining. Grounded theory coding consists of at least two phases: initial, which I explain here, and focused coding (see Chapter 6). During initial coding we study fragments of data – words, lines, segments, and incidents – closely for their analytic import. From time to time, we may adopt our participants' telling terms as codes. Initial coding continues the interaction that you shared with your participants while collecting data but brings you into an interactive analytic space.

In the coded interview excerpt below, a young woman called Teresa,[1] a graduate student in psychology, talked about having had surgery for anaplastic cancer of the thyroid gland when she was a 19-year-old voice major in college (see Wertz et al., 2011). Unlike other thyroid cancers, anaplastic cancer metastasizes rapidly and is life-threatening. Teresa had written an autobiographical account of her experience for an assignment in her graduate course in qualitative methods in which students were asked to write about an unfortunate event in their lives. Subsequently each student conducted and transcribed an interview with a classmate about this event. Teresa shared her autobiographical account and interview with all five researchers, who each analyzed the material. In both texts, Teresa talked of pursuing her lifelong goal of becoming an opera singer. She described herself as having been the extraordinary voice student that her professors and a growing public had already recognized. Teresa grew up in a family with a Filipino mother and a Venezuelan father with whom she often quarreled. In this interview excerpt, Teresa alluded to her troubled relationship with her father who neither supported her ambition nor believed she could earn a living as a singer.

BOX 5.1

Initial Grounded Theory Coding*

Examples of Codes	Initial Narrative Data to be Coded
	Could you talk about the ease… maybe… or difficulty… in the actual physical recovery?
Enduring recovery Remembering first instant of consciousness Measuring surgery in hours Finding unexpected (?) spread of tumor Explaining effects of anesthesia	It was horrible. I remember the instant I woke up from the surgery. And the surgery was supposed to take, maybe, three hours… it ended up taking something like six, maybe seven hours, because they didn't expect to find the spreading. I woke up…and… well, anesthesia has an interesting effect on people. I'd seen people come out of anesthesia before, and it's funny sometimes… people just start bawling and talking gibberish.
Waking up wailing Hearing a better voice Feeling jubilant about surgical results Being in pain	Naturally, I wake up and I just start wailing, crying. But I realize, first thing, that my voice is coming out much better than it had before surgery, so I thought, 'yeah, this is great!' The following weeks, I was in a lot of pain, primarily because of the nature of the surgery. For a thyroidectomy, there's a period of healing, of course, but my surgery was different
Increasing the excision Stopping the spreading tumor Being immobilized	because they had to go to the side of my neck where the tumor had begun to spread. As a result, I couldn't walk, could barely move… I was in bed for a good three weeks. I'm not the sort that can be bedridden easily. So I was mis-
Feeling miserable Being forced to stay with parents Wanting distance – dad	erable, and more unfortunate, I had to stay with my parents. My mother was fine… she doted on me a bit too much for my taste, but it was no surprise. But I could have done without my dad being there, and he was there plenty. And my
Continuing conflict complicating voice problems Externalizing her inability to speak coherently Experiencing altered speech	condition didn't mean we didn't argue, which just complicated things with my voice. Following the surgery, there was a notable inability to speak well for about a month, when my phonation was very definitively affected. Slowly, it started coming back here and there, but some-
Defining certain impairment Receiving no definitive explanation; not asking why; info withheld? Reciting possible loss	thing had definitely changed. I got everything checked, but no one could tell what changed. It's been theorized that the surgery was responsible for shifting some things around, so things were just going to be different from that point
Implying distress Defining permanent loss Experiencing forced loss *Voice and self merge*; losing valued self Acknowledging suffering	on. That was difficult… healing physically and coming to terms with the fact that things would have to be so different from then on… I wasn't even myself anymore after that. My voice was gone, so I was gone, and I'd never been anything but my voice. So yeah, that was really hard.

* From Charmaz (2011a, p. 173).

The interview excerpt in Box 5.1 includes my initial coding of Teresa's statements. The qualitative codes in the excerpt label what I saw as happening in the data fragments.

As Susan Leigh Star (2007) writes, 'a code sets up a relationship with your data, and with your respondents' (p. 80). Coding means naming segments of data with a label that simultaneously categorizes, summarizes, and accounts for each piece of data. With grounded theory coding, you move beyond concrete statements in the data to making analytic sense of stories, statements, and observations. We aim to make an interpretative rendering that begins with coding and illuminates studied life. If you concentrate on taking fragments of data apart and asking what meanings you glean from these fragments, you will move into analysis.

> Grounded theory coding is the process of defining what data are about.

> Coding means categorizing segments of data with a short name that simultaneously summarizes and accounts for each piece of data. Your codes show how you select, separate, and sort data and begin an analytic accounting of them.

For several years, I only knew Teresa through the two texts and formed my published analysis (Charmaz, 2011a) of her experience through coding these narratives as well as additional data from other people. Hence, I derived my codes from Teresa's texts but did not have first-hand knowledge of their contexts. When you collect first-hand data, however, you see the setting, observe interactions, witness research participants' non-verbal behaviour, and hear their voices as well as see written accounts. Analytic ideas may occur to you in the midst of an interview or during a moment in your ethnographic setting. If so, write a memo (see Chapter 7) about each idea so that you can develop and check it. Include the links between your idea and the empirical observation that spawned it. Novice researchers sometimes make the mistake of believing that only an interview transcription or a set of fieldnotes count as real data and ignore other events and cues that occur.

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The codes in Box 5.1 portray meanings and actions – and mark several questions that I had about Teresa's experience. Many of the codes are short. Note that the codes stick closely to the data, show actions, and indicate the progression of events from Teresa's point of view. The codes cover her descriptions of both her feelings and what happened to her as well as several additional codes of her explanations. Thus codes such as 'enduring recovery,' 'waking up wailing,' 'feeling jubilant about surgical results,' 'being in pain,' and 'feeling miserable' describe feelings. Codes such as 'being immobilized,' 'being forced to stay with parents,' and 'defining certain impairment' depict Teresa's story of what happened. I advise coding everything early in your research and see where it takes you as you proceed. These early codes can lead you to identify focused codes quickly.

Teresa moved from subjective description to an objective observation when she said 'there was a notable inability to speak well for about a month.' Her body then became an object of scrutiny which I called 'externalizing her inability to speak coherently.' I thought then that Teresa's statement and the code suggested important relationships – and tensions – between her body and self that might point to a line of analytic development.

My codes subsequently reveal intensity, momentum, and increased coherence as Teresa recounted changes and loss. Like many interviewees, she brings this story to its climax at the end of the passage. I viewed these last sentences, 'I wasn't even myself anymore after that. My voice was gone, so I was gone, and I'd never been anything but my voice,' as the most telling statements. Subsequently they formed the core of my analysis and identified one of the two major processes I analyzed, 'losing a valued self' (Charmaz, 2011a, p. 176–205). In keeping with Teresa's statements, my codes move from point to point. Note that her statements and, hence my coding, build tension as she learned of her condition. Codes such as 'experiencing altered speech,' 'receiving no definitive explanation,' 'reciting possible loss,' and 'defining permanent loss' begin to reveal processes involved in Teresa's experience of loss of her voice. The subsequent codes 'experiencing forced loss,' 'voice and self merge,' and 'losing a valued self' shaped the core of the analysis to follow.

Teresa articulated the depth of loss that numerous other people in my earlier studies had conveyed but seldom stated as explicitly. As a social psychologist who had studied the effects of chronic illness on the self, I was attuned to issues concerning loss of self (Charmaz, 1983b, 1991a; Swanson & Chenitz, 1993). Nonetheless, Teresa's statement testified to their significance for analyzing these data. 'Losing a valued self' reflects the data and thus in Glaser's (1978) terms[2] had earned its way into these codes and, by extension, into the subsequent analysis.

Consistent with a grounded theory emphasis on emergence, the questions I raise about these codes arise from my reading of the data rather than emanating from an earlier frame applied to them. I wondered if others knew that the cancer had spread before Teresa learned of it. Similarly, I also wondered if others had definitive information earlier than Teresa about the changes she had incurred from the surgery. Indicating such questions while coding gives you clues to consider later.

In the following pages, I lay out coding strategies for developing an analytic frame. Try them. See how they work for you. Grounded theory coding fosters studying action and processes, as you can see in the codes of Teresa's story. But first you may want more explanation of coding.

Coding gives you tools for interrogating, sorting, and synthesizing hundreds of pages of interviews, fieldnotes, documents, and other texts. Interrogating your data means that you take them apart and examine how these data are constituted. You create codes to explicate how people enact or respond to events, what meanings they hold, and how and why these actions and meanings evolved. Whether we have collected stories, scenes, or written statements, we study and define these materials to analyze what happened and what they might mean. Grounded theory coding is particularly useful for close examination and analysis of the data by breaking them into their components.

In grounded theory coding, we take segments of data apart, name them in concise terms, and propose an analytic handle to develop abstract ideas for interpreting each segment of data. As we code, we ask which theoretical categories might these statements indicate. Grounded theory coding is a powerful tool that can enable you to define what constitutes the data and to make implicit views, actions, and processes more visible. In brief, you begin to conceptualize what is happening in the data.

Next

The Logic of Grounded Theory Coding

About Coding

What does coding entail? How does grounded theory coding compare with other types of qualitative coding? Where does grounded theory coding take you?

Coding is the pivotal link between collecting data and developing an emergent theory to explain these data. Through coding, you *define* what is happening in the data and begin to grapple with what it means. The codes take form together as elements of a nascent theory that explains these data and directs further data-gathering. By careful attending to coding, you begin weaving two major threads in the fabric of grounded theory: generalizable theoretical statements that transcend specific times and places and contextual analyses of actions and events.

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Grounded theory coding generates the bones of your analysis. Theoretical centrality and integration will assemble these bones into a working skeleton. Thus, coding is more than a beginning; it shapes an analytic frame from which you build the analysis. As Star (2007, p. 84) notes, grounded theory codes are 'transitional objects.' Codes crystallize what we learn while studying our data. As transitional objects, they connect fragments of data with the analytic abstraction that we accord to them.

Conducting grounded theory coding involves you in at least two main phases: 1) an initial phase involving naming each word, line, or segment of data followed by 2) a focused, selective phase that uses the most significant or frequent initial codes to sort, synthesize, integrate, and organize large amounts of data. I aim to keep coding simple, direct, and spontaneous, but some researchers prefer more elaborate coding schemes. Should you wish to consider additional forms of early coding, Johnny Saldaña (2009) has both developed new ways of coding and compiled exemplars and explanations of established ways of coding in the most comprehensive treatment of qualitative coding available.

While engaged in initial coding, you mine early data for analytic ideas to pursue in further data collection and analysis. Initial coding entails a close reading of the data as indicated by my codes of Teresa's story. During initial coding, the goal is to remain open to all possible theoretical directions indicated by your readings of the data. Later, you use focused coding to pinpoint and develop the most salient codes and then put them to the test with large batches of data. Theoretical integration begins with focused coding and proceeds through all your subsequent analytic steps.

Your actual research will likely differ – at least somewhat – from what you may have planned earlier in a research or grant proposal (see, for example, Dunn, 2009). Your experiences in collecting and analyzing your data as you proceed shape that difference. We learn through what we find in our field settings and through studying the data we construct from them. Initial grounded theory coding guides our learning. Through it, we begin to make analytic sense of our data. How we make sense of it shapes the ensuing analysis. Careful attention to coding furthers our attempts to understand acts and accounts, scenes and sentiments, stories and silences from our research participants' view. We want to know what is happening in the setting, in people's lives, in documents, images, and texts, and in lines of our recorded data. Hence, we try to understand our participants' standpoints and situations, as well as their actions within the setting.

We learn through studying our data. Initial grounded theory coding guides our learning. Through it, we begin to make sense of our data. How we make sense of it shapes the ensuing analysis.

The logic of grounded theory coding differs from quantitative logic, which applies *preconceived* categories or codes to the data. The use of preconceived qualitative coding has also gained greater attention over the past decade from those who advocate application of extant theory or whose research proposals include completed coding books. In contrast, as the example above illustrates, grounded theorists *create* codes by defining what we see in the data. Codes emerge as you scrutinize your data and define meanings within it. Through this active coding, you interact with your data again and again and ask many different questions of them. As a result, coding may take you into unforeseen areas and new research questions.

Constructing Codes

We construct codes. Even concrete, descriptive codes show how we see the data. Language plays a crucial role in how and what we code. Most fundamentally, the empirical world does not appear to us in some natural state apart from human experience. Instead we know the empirical world through language and the actions we take toward it. In this sense, no researcher is neutral because language confers form and meaning on observed realities. Specific use of language reflects views and values. We share one language with colleagues and perhaps another with friends; we attribute meanings to specific terms and hold perspectives. Our codes arise from the languages, meanings, and perspectives through which we learn about the empirical world, particularly those of our research participants as well as our own. Coding impels us to make our participants' language problematic to render an analysis of it. Coding should inspire us to examine hidden assumptions in *our* use of language as well as that of our participants.

We *construct* our codes because we are actively naming data – even when we believe our codes form a perfect fit with actions and events in the studied world. We may think our codes capture the empirical reality. Yet it is *our* view: we choose the words that constitute our codes. Thus we define what we see as significant in the data and describe what we think is happening. Coding consists of this initial, shorthand defining and labeling; it results from a grounded theorist's actions and understandings. Nonetheless, the coding process is interactive. We interact with our participants and subsequently interact with them again many times over through studying their statements and observed actions and re-envisioning the scenes in which we know them. As we define our codes and perhaps later refine them, we try to understand participants' views and actions from their perspectives. These perspectives usually assume much more than what is immediately apparent. We must dig into our data to interpret participants' tacit meanings. Close attention to coding helps us to do that.

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This close attention to coding follows the first grounded theory decree: *study your emerging data* (Glaser, 1978). From the beginning, you may sense that the process of coding produces certain tensions – between analytic insights and described events, whether spoken accounts or written observations, between static topics and dynamic processes, and between participants' worlds and professionals' meanings. By studying your data and scrutinizing your codes you learn which of these tensions raise methodological and theoretical questions and which suggest substantive areas and leads for you to pursue.

Grounded theory coding need not be complex. By engaging in thorough coding early in the research process and comparing data and codes, the researcher can identify which codes to explore as tentative categories. In turn, selecting categories expedites inquiry because the researcher then uses these categories to sort large batches of data and simultaneously evaluate the relative usefulness of these tentative categories. This approach is particularly useful in social justice research projects that address pressing social issues and policies. Grounded theory coding preserves empirical detail and simultaneously moves the project toward completion.

Entering an Interactive Analytic Space

As my comments above indicate, grounded theory is not only an iterative, comparative method, but also an interactive method. It prompts you to keep interacting with your data. Grounded theory can bring you back to some research participants while going forward with fresh ideas to check with new participants (or with new texts or settings, depending on your sources of data). As you conduct grounded theory coding, you enter an interactive space that pulls you deeper into the data and keeps you involved with them far more than a casual reading fosters. In grounded theory coding, you *act* on your data and these actions sustain your involvement with them.

> As you conduct grounded theory coding, you enter an interactive space that pulls you deeper into the data and keeps you involved with them far more than a casual reading fosters. Being in this interactive space leads you to new analytic questions. The questions you ask merge the subjective with what appears to be objective as you grapple with understanding this world.

You may experience a sense of wonder about the world you are studying and feel awestruck about the privilege of learning about it. Being in this interactive space can challenge your earlier preconceptions and hunches. See what you can learn. Entering this interactive space means tolerating ambiguity as you grapple with making analytic sense of your data. It also means that you can see fragments of data in new ways and define new connections between them. Implicit meanings and actions grow visible.

Through grounded theory coding, we relive and re-view our earlier interactions with participants and subsequently interact with them again many times over. Our coding leads us to study their statements and observed actions and re-envision the scenes in which we know them. As we define our codes and perhaps later refine them, we try to understand participants' views and actions from their perspectives. These perspectives typically assume much more than what is immediately apparent. We must dig into our data to interpret participants' tacit meanings – and interact with these meanings again and again. Close attention to coding helps us to do that.

Being in this interactive space leads you to new analytic questions. The questions you ask merge the subjective with what appears to be objective as you contend with understanding this world. Your codes can shift, change, or transform your relationship with your data. Making comparisons between data and then data and codes advances and may transform your analytic understandings. As you learn about, know, and interpret your data, your codes capture both your involvement with these data and studied world and your analytic separation from them (see also Star, 2007).

Initial Coding

The Logic of Initial Coding

When grounded theorists conduct initial coding, we remain open to exploring whatever theoretical possibilities we can discern in the data. This initial step in coding moves us toward later decisions about defining our core conceptual categories. Through comparing data with data, we learn what our research participants view as problematic and begin to treat it analytically. During initial coding, we ask:

- **'What is this data a study of?' (Glaser, 1978, p. 57; Glaser & Strauss, 1967)**
- **What do the data suggest? Pronounce? Leave unsaid?**
- **From whose point of view?**

- **What theoretical category does this specific datum indicate? (Glaser, 1978)**

Initial coding should stick closely to the data. Try to see actions in each segment of data rather than applying pre-existing categories to the data. Attempt to code with words that reflect action. At first, invoking a language of action rather than of topics and themes may feel strange. Look closely at actions and, to the extent possible, code data *as* actions.

Coding for actions reduces tendencies to code for types of people. Coding people as types leads you to focus on individuals rather than what is happening in the data. Assigning types to people casts them with static labels. In effect, you make them one-dimensional although the behavior on which you based the label may represent only a small part of who they are and what they do. Both may change as events change. Thus, such coding freezes people in time and space and also erases or minimizes defining variation in the studied phenomenon.

In addition, coding for actions curbs our tendencies to make conceptual leaps and to adopt extant theories *before* we have done the necessary analytic work. Students often believe that they must rely on earlier concepts and invoke them before they begin coding to make their qualitative research legitimate. They adopt current ideas such as 'risk', 'habitus,' and 'discourses of power' to *apply* to their data rather than to initiate interrogating it. Similarly, other students and researchers may believe that they must apply a specific theorist to legitimize their work. 'I'm going to use Marx – or Mead – or Foucault – or Harding.'[3] Such applications can preclude *your* ideas from emerging as you code events. The openness of initial coding should spark your thinking and allow new ideas to emerge.

Earlier grounded theory rules prescribed conducting initial coding without having preconceived concepts in mind (Glaser, 1978, 1992). I agree with Glaser's approach of keeping initial coding open-ended but acknowledge that researchers hold prior ideas and skills. Dey's (1999, p. 251) oft-quoted criticism of Glaser and Strauss's (1967) stance still fits: 'There is a difference between an open mind and an empty head.' Try to remain open to seeing what you can learn while coding and where it can take you. Make efforts to learn and examine how your past influences the way you see the world and your data.

Sensitizing concepts can help you start to code your data. These concepts give you starting points for initiating your analysis but do not determine its content. Sensitizing concepts from symbolic interactionism include action, meaning, process, agency, situation, identity, and self. Social justice inquiry may lead to sensitizing concepts such as ideology, power, privilege, equity, and oppression, and to remaining alert to variation and difference. Glen Bowen (2006) chose 'empowerment' as one of several sensitizing concepts to conduct a small exploratory qualitative study of community improvement projects that the Jamaica Social Investment Fund (JSIF) had supported. As he engaged in line-by-line coding, Bowen found that empowerment was not a directly useful concept and dropped it. His initial interest in empowerment, however, did foster pursuing a fruitful direction. It led him to seek nuanced data and to construct useful codes. He writes: 'In general, the JSIF-funded projects were more enabling than empowering' (p. 20). Among his codes, Bowen defined 'creating an enabling environment' as a significant consequence of citizen involvement in the process of enacting an improvement project.

Initial codes are provisional, comparative, and grounded in the data. They are provisional because you aim to remain open to other analytic possibilities and create codes that best fit the data you have. You progressively follow up on codes that indicate that they fit the data. Then you gather data to explore and fill out these codes. In team research, several individuals may code data separately and then compare and combine their codes to evaluate their fit and usefulness. Might one team member come up with different codes than other members? Yes, our perspectives, social locations, and personal and professional experiences affect how we code. Thus, team researchers can scrutinize how differences among team members may generate new insights, rather than dismiss a colleague's codes that differ.

Initial grounded theory coding can prompt you to see areas in which you lack needed data. Realizing that your data have gaps – or holes – is part of the analytic process. It is inevitable when you adopt an emergent method of conducting research.[4] After all, making 'discoveries' about the worlds you study and pursuing these discoveries to construct an analysis is what grounded theory is about. Such discoveries reflect what you learn and how you conceptualize it. The advantage of grounded theory strategies is that you may learn about gaps and holes in your data from the earliest stages of research. Then you can locate sources of needed data and gather them. Hence, simultaneous data collection and analysis can help you go further and deeper into the research problem as well as engage in developing categories.

Codes are also provisional in the sense that you may reword them to improve their fit with data. Part of the fit is the degree to which they capture and condense meanings and actions. Compelling codes capture the phenomenon and grab the reader. Speed and spontaneity help in initial coding. Working quickly can spark your thinking and spawn a fresh view of the data. Some codes fit the data and grab the reader immediately. You can revise others to improve the fit.

In the example below, Bonnie Presley talked of her difficulty in telling her adult daughter, Amy, about her recent crisis. Amy now lived in the same area, although their phone calls and visits remained sporadic. Years before, Amy could not understand how the mother she had known as a fitness buff could have become so sedentary. Bonnie's youthful physical appearance belied her health status because her symptoms remained invisible to an untrained eye. In the early years of her illness, Bonnie had long found it difficult to tell Amy about her illness and its seriousness. Amy had moved away before Bonnie first became ill and Bonnie had either understated what was happening or avoided telling Amy. Bonnie recounted her realization about how she had told Amy the news this time, as the data in Box 5.2 shows.

My original code of the first line of Bonnie Presley's story was 'receiving news indirectly.' It condensed the statement but the neutral wording drained the incident of its intensity and importance. Changing the code to 'receiving second-hand news' suggested the reduced value of the news, implied the receiver's diminished status, and alluded to her angry response. You might wonder if it is legitimate to accept Bonnie's account of Amy's response without interviewing Amy. Yes, it is, because

I focused on how chronically ill people dealt with news about their conditions. 'Receiving second-hand news' emerged as a consequence of the amount of interaction, relative invisibility of illness and disability, and past patterns of limited disclosure or non-disclosure.

BOX 5.2

Grounded Theory Initial Coding Example

<u>Initial codes</u>	<u>Interview statement</u>
Receiving second-hand news; Being left out; Accusing mother of repeated not telling; (questioning ethical stance?) Being confronted Facing self and identity questions; Demanding self-disclosure and information	She found out from Linda that I was, had been in bed for days and she called me up, 'You never tell me, and I have to find out from Linda,' and 'Why don't you tell me who you are and what's going on and ...'
Experiencing escalating pain Expecting to manage pain Inability to control pain	Well, I don't know how long after that, but that Saturday the pain started right here and it, throughout the day it got worse and worse and worse. And she – I kept thinking that, well, I can deal with this, so I took some kind of a pain pill and nothing helped.
Rapid worsening of pain	And that was about one in the afternoon. Well, it got worse and worse so that every time I took a breath the pain was horrible,
Having excruciating pain Becoming frightened Foreseeing breathing crisis Breaking the news; informing daughter of plan Explaining projected treatment	so by seven, eight o'clock that night, I was scared because I knew that if it got any worse I wasn't going to be able to breathe. So I called her and then I told her what was going on, that I was going to be driven to the doctor because they were going to try giving me shots of zylocain or something to try to locate a point to where maybe it would go in there and numb the pain for me so that I could breathe.
Having access for making contact Leaving follow-up contact open-ended No follow-up Ascertaining the time between contacts Explaining lack of disclosure	Well, I called her and told her this. And I have a car phone. She says, 'Well, Mom I'll call you later or you call me.' Well, I didn't call her; she didn't call me. That was Saturday night. She didn't call me until – she called me about noon on Monday, and I finally said, 'Well look, this is why I don't tell you,
Accusing daughter of not caring Expressing hurt; Assuming lack of caring; Making negative inferences (of a moral lapse?) Accounting for not telling Sounding fine Questioning daughter's expectations	because when I told you Saturday night, you never called, you didn't care or anything and it really hurt my feelings. So that's why I don't tell you when I have this going on.' And she said to me, 'Well, Mom, you sounded perfectly fine.' And I said, 'Well, what do you expect me to do, become an emotional wreck or something?'
Explaining need for emotional control Seeing life-threatening risk of losing control Teaching that mode of telling does not reflect state of being	I said, 'I have to keep everything still and quiet in me in order to control, because if I went into emotional frenzy, I would have not been able to breathe,' you know. So she started really trying to understand that just because I was scared to death, I was in horrible pain,
Sounding like a 'normal' mom	but when I called her, I guess I was just a normal mom.

At this point a key difference between grounded theory and other forms of qualitative research on accuracy becomes apparent. Recall our stance on the theoretical plausibility of data. Similarly, grounded theorists aim to code for possibilities suggested *by* the data rather than ensuring complete accuracy *of* the data. This approach helps you to define the range of variation of your studied process or phenomenon and provides leads for conceptualizing it further and checking your ideas with other data.

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Comparing incidents of the same order between data spurs you to think analytically about them. Bonnie Presley revealed a reluctance to tell her daughter, delayed in telling her, and imparted difficult news in a matter-of-fact manner. Yet, from time to time over the years, she and Amy had talked about their problems in giving and getting news about Bonnie's illness. Because Bonnie no longer had much contact with her own mother, dilemmas of disclosure did not arise with her. No disclosures occurred. Bonnie's grandmother, of whom she was very fond, had partly raised her. Bonnie protected her grandmother from worry by treating her situation lightly and by minimizing the implications of her symptoms. My data included other cases of intergenerational tensions. Several other single women I studied who had no children and few close family ties had conflicted relationships with their aging mothers. As geographical and emotional distance increased, these women correspondingly curtailed sharing their news. From the data and brief descriptions above, avoiding disclosure, delaying disclosure, and controlling information all emerged as salient codes.

Picking up general terms from an interview such as 'experience' or 'event' and calling them codes tells you little about the participant's meaning or action. If general terms seem significant, qualify them. Make your codes fit the data you have rather than forcing the data to fit them.

A code for coding:

- **Remain open**
- **Stay close to the data**
- **Keep your codes simple and precise**
- **Construct short codes**
- **Preserve actions**
- **Compare data with data**
- **Move quickly through the data.**

In sum, remain open to what the material suggests and stay close to it. Keep your codes short, simple, active, and analytic. The first two guidelines above reflect your stance toward coding. The remaining guidelines suggest how to do coding.

Coding for Topics and Themes vs. Coding with Gerunds

Glaser (1978) shows how coding with gerunds helps you detect processes and stick to the data. Think of the difference in imagery between the following gerunds and their noun forms: describing versus description, stating versus statement, and leading versus leader. We gain a strong sense of action and sequence with gerunds. The nouns turn these actions into topics. Staying close to the data and, when possible, starting from the words and actions of your respondents, preserves the fluidity of their experience and gives you new ways of looking at it. These steps encourage you to begin analysis from their perspective. That is the point. If you ignore, gloss over, or leap beyond participants' meanings and actions, your grounded theory will likely reflect an outsider's, rather than an insider's view. Outsiders often import an alien professional language to describe the phenomenon. If your data are thin and if you do not push hard in coding, you may mistake routine rationales for analytic insights. Thus, accepting participants' orchestrated impressions at face value can lead to outsider analyses.

Line-by-line coding, the initial grounded theory coding with gerunds, is a *heuristic* device to bring the researcher into the data, interact with them, and study each fragment of them. This type of coding helps to define implicit meanings and actions, gives researchers directions to explore, spurs making comparisons between data, and suggests emergent links between processes in the data to pursue and check.

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> researcher into the data, interact with them, and study each fragment of them. This type of coding helps to define implicit meanings and actions, gives researchers directions to explore, spurs making comparisons between data, and suggests emergent links between processes in the data to pursue and check.

The data excerpts in Boxes 5.3 and 5.4 tell the story of a middle-aged woman with lupus erythematosus whose friends rushed her to a doctor to reassess her medications during a medical crisis. These medications can cause multiple side-effects including confusion, depression, blurred vision, and inappropriate emotional responses. After two hospital transfers, this woman's medical crisis became redefined as a psychiatric crisis. Subsequently, her claims of having physical symptoms were unacknowledged and her requests for lupus medications went unheeded. She aroused the doctor's ire when he discovered another patient helping her complete the detailed personal intake survey. Getting help with the survey broke all hospital rules of confidentiality. However, this woman's vision problems meant that she could not read the questions and so she just filled circles randomly after her doctor forbade her from having help. Her actions in one incident after another made sense given her situation, but fit neither the hospital protocol nor the logic of her treatment program.

Nevertheless, this woman attempted to present her views and to become her own advocate while her physical illness worsened. If the psychiatrist was either unaware of her medical history or ignored it, he could have invoked the same incidents as justifying his diagnosis and treatment. In this case, the grounded theory codes chronicle this woman's progressive loss of control over her life and her illness. Thus the initial codes in the excerpt become the details substantiating a more general code, 'resisting spiraling powerlessness.'

Comparing the codes in Boxes 5.3 and 5.4 demonstrates that grounded theory codes preserve the character of the data, provide a precise handle on the material, and point to places that need further elucidation. Coding gives the researcher leads to pursue in subsequent data collection.

Note the difference between coding for topics as contrasted with grounded theory coding for actions. General qualitative coding identifies topics about which the researcher can write; the researcher may use these topics as areas to sort and synthesize the material. Line-by-line grounded theory coding goes deeper into the studied phenomenon and attempts to explicate it.

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BOX 5.3

Initial Coding for Topics and Themes

Examples of Codes	Narrative Data to be Coded
Friends' support Hospitalization Conflict with doctor Hospital transfer Loss of choice of doctor Conflict with doctor Physician control Threats Powerlessness Lack of physical care	P: They called the clinic to see if they could see me, if they would reevaluate some of my meds and stuff, and they said, 'Oh yeah.' When I got there they decided that they were going to put me in, put me away or whatever. And I ended up with a really bad doctor. Really bad. I even brought charges against him, but I lost. I: What did he do? P: They put me in this one place, then the next day they sent me over to West Valley [hospital 60 miles away], and they didn't have any female doctors there, they only had male, so you didn't have a choice, and you get one and that's who you get the whole time you're there. For some reason he just took a disliking, I guess, and I tried to tell him about some of the problems I had with my Lupus and stuff, and angered him. [He had ordered her to take off her dark glasses.] And I wore [dark] glasses all the time and I tried to tell him, you know, that if he would turn off the fluorescent lights, I would take off the glasses. And he felt I was just being stubborn. I gave him the name and number of my doctor that makes the glasses and he just ripped it up in front of me and threw it away. And you have to go to group sessions all the time while you're there. I went, I just didn't speak to anybody. But I went. I went to everything they said I had to. And everyday he'd say that he was going to lock me up again. After ten days he did. He called me in to this little room and there were these two big guys there and they grabbed me and put me on a gurney and tied me down. And he sent me to a lockup ward, wouldn't let me make a call or do anything. And my potassium was down really bad and, I mean my whole – I was sick. They weren't giving me the pills that I needed and he just wouldn't even acknowledge that I had Lupus. It was bad.

BOX 5.4

Grounded Theory Coding in Comparison with Thematic Coding

Examples of Codes	Initial Narrative Data to be Coded
Receiving friends' help in seeking care Requesting regimen reevaluation Gaining medical access Being admitted to hospital Getting a 'bad' doctor Taking action against MD Being sent away Preferring a female MD Losing choice; dwindling control Getting stuck with MD Accounting for MD's behavior Trying to gain a voice – explaining symptoms Remaining unheard Asserting self Attempting to bargain Being misjudged Countering the judgment Offering evidence Being discounted Facing forced attendance Maintaining silence 'Following' orders Receiving daily threats MD acting on threat Being overpowered Experiencing physical constraints Experiencing immediate loss of control Witnessing one's deterioration Being denied medication MD rejecting illness claims	P: They [her friends] called the clinic to see if they could see me, if they would reevaluate some of my meds and stuff, and they said, 'Oh yeah.' When I got there they decided that they were going to put me in, put me away or whatever. And I ended up with a really bad doctor. Really bad. I even brought charges against him, but I lost. I: What did he do? P: They put me in this one place, then the next day they sent me over to West Valley [hospital 60 miles away], and they didn't have any female doctors there, they only had male, so you didn't have a choice, and you get one and that's who you get the whole time you're there. For some reason he just took a disliking, I guess, and I tried to tell him about some of the problems I had with my Lupus and stuff, and angered him. [He had ordered her to take off her dark glasses.] And I wore [dark] glasses all the time [because of her photosensitivity] and I tried to tell him, you know, that if he would turn off the fluorescent lights, I would take off the glasses. And he felt I was just being stubborn. I gave him the name and number of my doctor that makes the glasses and he just ripped it up in front of me and threw it away. And you have to go to group sessions all the time while you're there. I went, I just didn't speak to anybody. But I went. I went to everything they said I had to. And everyday he'd say that he was going to lock me up again. After ten days he did. He called me in to this little room and there were these two big guys there and they grabbed me and put me on a gurney and tied me down. And he sent me to a lockup ward, wouldn't let me make a call or do anything. And my potassium was down really bad and, I mean my whole – I was sick. They weren't giving me the pills that I needed and he just wouldn't even acknowledge that I had Lupus. It was bad.

Line-by-line coding gives researchers more directions to consider and already suggests emergent links between processes in the data. The codes above are quite concrete and close to the data but indicate spiraling events during which P's loss of control escalated. The codes also illuminate the situations in which these events occur; readers gain a sense of what is happening in this statement and how it happens. The analytic level of the grounded theory codes ranges from describing a fragment of data, such as 'trying to gain a voice' and 'remaining unheard,' to potential analytic categories such as 'experiencing immediate loss of control.' When your codes are concrete, examine them and ask what larger analytic story they indicate.

If coding in gerunds is so fruitful, why don't more researchers use them? In my view, the English language favors thinking in structures, topics, and themes rather than thinking in actions and processes. In addition, Strauss and Corbin's (1990, 1998) books have instructed thousands of researchers but emphasize gerunds less than Glaser (1978, 1998) and I (1990, 2008c) do. Many grounded theorists begin with the open-minded stance that Glaser and Strauss (1967) first recommended but do not use gerunds (see, for example, Allen, 2011; Andrade, 2009; Porr in Stern & Porr, 2011, pp. 82–83). Furthermore, other researchers' coding decisions lead them to adopt and adapt grounded theory strategies to manage qualitative data rather than to study processes and construct theories. Yet coding with gerunds and studying processes enable you to discern implicit connections – and, simultaneously, give you control over your data and emerging analysis.

Initial Coding Practices

Word-by-Word Coding

The size of the unit of data to code matters. Some grounded theorists conduct nuanced coding and move through their data word by word. Those with interests in phenomenology may find word-by-word coding to be a complementary coding strategy. This approach may also be particularly helpful when working with documents or certain types of ephemera, such as data from internet blogs. Word-by-word analysis forces you to attend to images and meanings. You may attend to the structure and flow of words, and how both affect the sense you make of them, as well as their specific content.

Line-by-Line Coding

For many grounded theorists, line-by-line coding is the first step in coding. Line-by-line coding means naming each line of your written data (Glaser, 1978). Coding every line may seem like an arbitrary exercise because not every line contains a complete sentence and not every sentence may appear to be important.[5] Nevertheless, it can be an enormously useful tool. Ideas will occur to you that had escaped your attention when reading data for a general thematic analysis.

Line-by-line coding works particularly well with detailed data about fundamental empirical problems or processes, whether these data consist of interviews, observations, documents, or ethnographies and autobiographies. For example, pretend that you plan to study how people end an addiction to prescription drugs. You have identified an area to explore about which you would hear stories in interviews, support groups, and written personal accounts that take on vivid meanings when studied line by line.

Detailed observations of people, actions, and settings revealing their everyday life as well as visibly *compelling* and *consequential* scenes and actions lend themselves to line-by-line coding. Generalized observations afford you little such substance to code. Line-by-line coding encourages you to see otherwise undetected patterns in everyday life. Line-by-line coding enables you to take compelling events apart and analyze what constitutes them and how they occurred.

Fresh data and line-by-line coding prompt you to remain open to the data and to see nuances in them. When you code early in-depth interview data, you gain a close look at what participants say and, likely, struggle with. This type of coding can help you to identify implicit concerns as well as explicit statements. Engaging in line-by-line coding helps you to refocus later interviews. The following flexible strategies help you code:

- **Breaking the data up into their component parts or properties**
- **Defining the actions on which they rest**
- **Looking for tacit assumptions**
- **Explicating implicit actions and meanings**
- **Crystallizing the significance of the points**
- **Comparing data with data**
- **Identifying gaps in the data.**

By using these strategies flexibly and following leads in your data, coding leads to developing theoretical categories, some of which you may define in your initial codes. Stick with what you define in your data. Build your analysis step-by-step from the ground up without taking off on theoretical flights of fancy. Having a credible amount of data that speaks to your research topic further strengthens the foundation of your study.

Your research participants' actions and statements teach you about their worlds, albeit sometimes in ways they may not anticipate. Studying your data through line-by-line coding sparks new ideas for you to pursue. Hence, the grounded theory method itself contains correctives that reduce the likelihood that researchers merely superimpose their preconceived notions on the data. Line-by-line coding provides an early corrective of this type.

In the example of line-by-line coding in Box 5.5 on the following page, Kris Macomber describes how her data had revealed women advocates' multiple and sometimes conflicting viewpoints about men's involvement in the movement and then she shows us how Ruby expressed these multiple viewpoints. Note that Macomber uses 'conveying multiple viewpoints' as a code three times in this excerpt, which affirms its significance. Her codes simultaneously synthesize and analyze key statements in the interview excerpt.

Initial codes often range widely across a variety of topics. Because even a short statement or excerpt may address several points, it could illustrate several different categories. I could use the excerpt from Bonnie Presley's interview in Box 5.2 to show how avoiding disclosure serves to control identity. I could also use it either to show how a seriously ill individual creates strategies to control crises or why he or she withdraws from significant relationships. Having multiple interviews of the same individuals allows me to see how social and emotional isolation begins and progresses.

BOX 5.5

Kris Macomber's Initial Line-by-Line Coding

Given that the Violence Against Women Movement is a historically feminist women's movement, I wanted to know how women advocates felt about men's increasing involvement. Based on what I learned about the tensions men's involvement was creating within the movement, I expected to find that some advocates would support men's involvement, while others would resist it. During interviews, I asked women advocates, 'How do you feel about men's increasing involvement in the movement?' Through continuous analytic coding I discovered that women's reactions to men's involvement were not nearly as straightforward as I expected.

During initial coding, I coded data pieces with codes like 'conveying multiple viewpoints,' and 'critiquing male privilege.' Early on, these codes captured how individual women were making sense of men's increasing involvement, including their concerns about men's impact on the movement. My initial coding also identified the complexity of women's reactions to men's involvement. See below.

Initial Coding

Conveying multiple viewpoints Distinguishing between viewpoints Attributing responsibility to men Evaluating men's increasing involvement Distinguishing between viewpoints Conveying multiple viewpoints; claiming men have elevated status in movement. Critiquing reactions to men's involvement Claiming men activists are romanticized Claiming inequity; claiming sexism Identifying and critiquing male privilege Perceiving men as 'elevated,' 'higher' Comparing men's contributions to women's Conveying multiple viewpoints Critiquing reactions to men, casting men as undeserving; comparing responses to men with responses to women.	Excerpt 1 Ruby, age 31, rape prevention specialist: I have a few different trains of thought about men's involvement. One is, why in the hell wouldn't they be involved? The overwhelming majority of perpetrators are men. It's also about time more men are doing this work. So, that's one thought that I have – that they should be involved. But, at the same time, I'm sure not going to put anybody on a pedestal. I've seen that happen way too often... 'Oh wow, look at this man. And he's involved in the anti-sexual violence movement and isn't that great!' 'Wow, he's dreamy!' There's all this sexist stuff in this movement, 'Oh well, it's a man and men are better, so we will elevate them on this great pedestal...' Even though what they're saying is the same exact thing that Ida B. Wells said back in like 1824. I think it's great that men are involved because they need to be, but I do not agree with putting men who are involved on a pedestal because women certainly haven't been.

The logic of 'discovery' becomes evident as you begin to code data.

Line-by-line coding prompts you to look at the data anew. Compare what you see when you read a set of fieldnotes or an interview as an entire narrative with what you gain when you do word-by-word, line-by-line, or incident-with-incident coding on the same document. Entire narratives may net several major themes. Word-by-word, line-by-line, segment-by-segment, and incident-with-incident coding may generate a range of ideas and information. Therefore, you 'discover' ideas on which you can build.

> Line-by-line coding prompts you to look at the data anew. Initial codes help you to separate data into categories and to see processes. Line-by-line coding frees you from becoming so immersed in your research participants' world-views that you accept them without question.

Initial codes help you to separate data into categories and to see processes. Line-by-line coding frees you from becoming so immersed in your research participants' world-views that you accept them without question. If you had already accepted the same assumptions inhering in your participants' world-views, then you risk reproducing them in your research. This problem not uncommonly occurs when researchers from professional fields study members from their respective professions. In either case, you fail to look at your data critically and analytically – and to seek data that question the assumptions undergirding the views and practices you study. Being critical about your data does not necessarily mean being critical of your research participants. Instead, being critical forces asking *yourself* questions about your data. These questions help you to see actions and to identify significant processes. Such questions include:

- **What process(es) is at issue here? How can I define it?**
- **How does this process develop?**
- **How does the research participant(s) act while involved in this process?**
- **What does the research participant(s) profess to think and feel while involved in this process? What might his or her observed behavior indicate?**
- **When, why, and how does the process change?**
- **What are the consequences of the process?**

Through coding each line of data, you gain insights about what kinds of data to collect next. Thus, you distill data and direct further inquiry early in the data collection. Line-by-line coding gives you leads to pursue. If, for example, you identify an important process in your fifteenth interview, then you can return to earlier respondents and see if that process explains events and experiences in their lives. If you cannot return, you can seek new respondents who can illuminate this process. Hence, your data collection becomes more focused, as does your coding.

If perchance your line-by-line codes remain at a mundane level, then *code these codes*. Ask yourself, what larger analytic story do these codes suggest? What process do they suggest? By coding the codes, you push yourself to look for patterns and think more analytically – and you keep interacting with your data and codes.

> If perchance your line-by-line codes remain at a mundane level, then *code these codes*. Ask yourself, what

> larger analytic story do these codes suggest? What process do they indicate? By coding the codes, you push yourself to look for patterns and think more analytically – and you keep interacting with your data and codes.

Coding Incident with Incident

Whether or not you conduct line-by-line coding depends on the type of data you have collected, their level of abstraction, the stage of the research process, and your purpose for collecting these data. Grounded theorists often conduct a close cousin of line-by-line coding through a comparative study of incidents. Here you compare incident with incident, then as your ideas take hold, compare incidents to your conceptualization of incidents coded earlier. That way you can identify properties of your emerging concept. In Box 5.6, I compare two incidents in which each young woman learns of her physical loss, which I have coded as 'comprehending the ominous moment,' to mirror thoughts and feelings that each research participant expressed. The two incidents share the properties of a rapid onset and of each person's sudden realization of its meaning.

We can code and compare these incidents in several ways. First, we can look at the context of each incident and compare them.

Teresa had had visible changes and thus limited forewarning that something was amiss, although she did not know what it portended. Gail's injury, in contrast, occurred without warning. When comparing the content of these two incidents, the codes outline the sequence of becoming aware of what was happening, portray experiencing these moments, and reveal what each woman recounted thinking and feeling. Teresa learned of the seriousness of her condition from another person, the surgeon. Gail learned of her injury through experiencing the speed of her descent, hearing the sound of her elbow tearing, and feeling her elbow and contorted arm. Box 5.7 shows how we can compare these two incidents to discern their properties.

I used the same excerpts to compare the two incidents, which I coded as 'comprehending the ominous moment.' I saw the incidents as ominous moments in congruence with the women's statements and ferreted out the properties of these moments. By coding in this way, I could either remain at the level of the experience or embark on an abstract analysis of relationship between time, identity, and awareness. The properties give me direction for coding similar incidents and for writing memos to expand these ideas.

You can use a similar logic with observational data. Making comparisons between incidents may work better than word-by-word or line-by-line coding, in part because fieldnotes already contain a logic and point of view that you have given them. After all, fieldnotes consist of your words, your framing of events, and your way of recording these events to make them comprehensible (see, for example, Charmaz & Mitchell, 2001). Compare incident with incident. Isolated concrete, behavioristic descriptions of people's mundane actions may not be amenable to fruitful line-by-line coding, particularly when you observed a scene but neither interacted in it nor gained a sense of its context, its participants, or their intentions for their actions.[6]

When, however, you participate in the scene or setting and record detailed fieldnotes, then your initial codes may be more plentiful.

BOX 5.6

Comparing Incident with Incident Codes: Comprehending the Ominous Moment

Excerpt 1: Teresa on learning she could lose her voice		Excerpt 2: Gail on her injury during gymnastics practice	
I took it all in very methodically, as though we were talking about someone else entirely that he'd be cutting into the next morning.	Separating bad news from self	All of a sudden, with the blink of an eye, I feel my body coming down quickly… instantly… toward the ground. I'm going fast. I'm almost head first going like a torpedo on to the mat underneath the bar.	Defining fateful speed and direction
After that was done, he leaned back in his chair and asked me if I had any questions, and I didn't. Then, perhaps in an effort to make small talk, he asked, 'So, you're a college student… what's your major?' I told him it was vocal performance, and his face went white. He looked grimmer now than he had at any point in our conversation.	Speed – quickening of cues	Somehow I managed to miss the bar. I was so high, but too far away from the high bar to catch it. I'm coming down fast. Even though it was so fast, I felt that moment take forever. All of a sudden I hear a crack. Or was it a tear? It sounded like the Velcro that holds the mats together ripping apart.	Feeling the moment last Hearing the impact
'Look,' he said very gently, 'because of where this thing is and what we're going to have to do, there's a chance you won't be able to even speak the same way again. You may not be singing anymore after this.'	Forecasting grim Future Imparting ominous news	I almost turned to see what it was. Wait. Something feels funny. Wait. Something doesn't feel right. I was on the floor kneeling down underneath the high bar.	Experiencing gap between sound and feeling
I froze. I couldn't breathe, couldn't move, couldn't even blink. I felt like I	Shortening of time Immediate awareness	I feel my right elbow with my left hand. Something feels very, very wrong. There was no elbow anymore, my arm was contorted. I couldn't feel that bony part of my arm. It was bent the wrong way. I panicked. That Velcro sound was from my elbow? I'm holding my	Feeling her injury Sensing horror

(Continued)

had just been shot. My gut had locked up like I'd been punched in it. My mouth went dry and my fingers, which had been fumbling with a pen, were suddenly cold and numb.	Collapsed future	arm, feeling the new bend created by the fall to the floor. Then it hits me. Look at what happened to me, in a split second. I thought about my competitive season… going down the drain. I thought about sitting out all those meets…again. I thought about the doctor.	Realizing what happened Foretelling the sequel
Apparently picking up on my shock, the surgeon smiled a little. 'We're going to save your life, though. That's what counts. And you know what? The other surgeon working with me is a voice guy. We're going to do everything we can not to be too intrusive.'	Softening bad news	I thought about surgery. I panicked more when I thought about surgery. I remember the shock. When I felt my elbow, I said 'Oh my God!' 'Oh my God!' in panic and disbelief that something so intense could happen in a split second.	Becoming panicked
I started to breathe a little, very little, and I felt myself trembling. I tried to say something meaningful, expressive… all that I could manage was, 'Man… I was actually pretty good.' Then, all of me let loose. I was sobbing, but there was no sound; just a torrent of tears, and the hiss of crying from my open mouth, pushing through the pressure from the accursed mass.	Grieving profound loss	Then, as it all started to sink in and the panic came over me, I kept saying, 'No!! No!! No!!', first in denial and passionate, then through sobs and a feeling of defeat and frustration.	Feeling defeated and frustrated

BOX 5.7

Comparing Properties in Incident with Incident Coding:

<u>Comprehending the Ominous Moment</u>

Excerpt 1: Teresa on learning she could lose her voice

I took it all in very methodically, as though we were talking about someone else entirely that he'd [the surgeon] be cutting into the next morning. After that was done, he leaned back in his chair and asked me if I had any questions, and I didn't. Then, perhaps in an effort to make small talk, he asked, 'So, you're a college student… what's your major?' I told him it was vocal performance, and his face went white. He looked grimmer now than he had at any point in our conversation. 'Look,' he said very gently, 'because of where this thing is and what we're going to have to do, there's a chance you won't be able to even speak the same way again. You may not be singing anymore after this.'

I froze. I couldn't breathe, couldn't move, couldn't even blink. I felt like I had just been shot. My gut had locked up like I'd been punched in it. My mouth went dry and my fingers, which had been fumbling with a pen, were suddenly cold and numb. Apparently picking up on my shock, the surgeon smiled a little. 'We're going to save your life, though. That's what counts. And you know what? The other surgeon working with me is a voice guy. We're going to do everything we can not to be too intrusive.' I started to breathe a little, very little, and I felt myself trembling. I tried to say something meaningful, expressive… all that I could manage was, 'Man… I was actually pretty good.'

Then, all of me let loose. I was sobbing, but there was no sound; just a torrent of tears, and the hiss of crying from my open mouth, pushing through the pressure from the accursed mass.

From Wertz et al. (2011, pp. 105–106)

Excerpt 2: Gail on her injury during gymnastics practice

All of a sudden, with the blink of an eye, I feel my body coming down quickly… instantly… toward the ground. I'm going fast. I'm almost head first going like a torpedo on to the mat underneath the bar. Somehow I managed to miss the bar. I was so high, but too far away from the high bar to catch it. I'm coming down fast. Even though it was so fast, I felt that moment take forever. All of a sudden I hear a crack. Or was it a tear? It sounded like the Velcro that holds the mats together ripping apart. I almost turned to see what it was. Wait. Something feels funny. Wait. Something doesn't feel right. I was on the floor kneeling down underneath the high bar. I feel my right elbow with my left hand. Something feels very, very wrong. There was no elbow anymore, my arm was contorted. I couldn't feel that bony part of my arm. It was bent the wrong way. I panicked. That Velcro sound was from my elbow? I'm holding my arm, feeling the new bend created by the fall to the floor. Then it hits me. Look at what happened to me, in a split second. I thought about my competitive season… going down the drain. I thought about sitting out all those meets…again. I thought about the doctor. I thought about surgery. I panicked more when I thought about surgery. I remember the shock. When I felt my elbow, I said 'Oh my God!' 'Oh my God!' in panic and disbelief that something so intense could happen in a split second. Then, as it all started to sink in and the panic came over me, I kept saying, 'No!! No!! No!!', first in denial and passionate, then through sobs and a feeling of defeat and frustration.

From Wertz et al. (2011, pp. 404–405)

Properties of 'Comprehending the Ominous Moment'

1 Perceiving Rapid Successive Cues – *Speed and Timing*
2 Realizing the Meaning of Cues – *Instantaneous Awareness Despite Shock*
3 Experiencing Pivotal Point of Self – *Defined Turning Point*
4 Shattering the self – *Incident Strikes Core of Self and Identity*
5 Immediate Understanding of Consequences – *Foretells Future*

Nonetheless, detailed observations alone do not guarantee creating an insightful theoretical analysis although they might generate excellent description. The mode of analysis matters. Comparative methods help you to see and make sense of observations in new, analytic ways. Conducting a line-by-line coding of one observation after another of people's mundane actions in a public place may not spark fresh ideas. If so, try making comparisons between these observations. You will likely discern clues to follow if not have immediate ideas. If the people you study bring you into their world, for example, you may record all kinds of incidents in anecdotes, conversations, and observations in your fieldnotes that abound with meaning (recall Gongaware's (2012, pp. 10–11) comments discussed in Chapter 2). You may see first-hand how your participants manage daily life without them telling you – and you may learn much more.

The more unproblematic – that is, routine, familiar, and ordinary – observed events seem to you, the more problematic creating an original conceptual analysis of them will be. Breaking through the ordinariness of routine events takes effort. To gain analytic insights from observations of routine actions in ordinary settings, first compare and code similar events. Then you may define subtle patterns and significant processes. Later, comparing *dissimilar* events may give you further insights.

Using Comparative Methods

Whatever unit of data you begin coding in grounded theory, you use *constant comparative methods* (Glaser & Strauss, 1967) to establish analytic distinctions – and thus make comparisons at each level of analytic work. At first, you compare data with data to find similarities and differences. For example, compare interview statements and incidents within the same interview and compare statements and incidents in different interviews. Making sequential comparisons helps. Compare data in earlier and later interviews of the same individual(s) or compare observations of events at different times and places. When you conduct observations of a routine activity, compare what happens on one day with the same activity on subsequent days.

If your codes define another view of a process, action or belief than your respondent(s) holds, note that. Your observations and ideas matter. Don't dismiss your own ideas if they differ from statements in the data. These ideas may rest on covert meanings and actions that have not entirely surfaced yet. Such intuitions form another set of ideas to check. Our task is to make analytic sense of the material, which may challenge taken-for-granted understandings, and grounded theory strategies lead you to remain engaged in comparative analysis to test your ideas.

What you see in your data relies in part upon your prior perspectives. Rather than seeing your perspectives as truth, try to see them as representing one view among many. That way, you may gain more awareness of the concepts that you employ and might impose on your data. To illustrate, you might already possess a repertoire of psychological concepts through which you ordinarily explain behavior. Invoking these concepts in your codes can lead you to prejudge what is happening. Try to avoid assuming that respondents, for example, repress or deny significant 'facts' about their lives. Instead, look for how they understand their situations before you judge their attitudes and actions through your own assumptions. Seeing the world through their eyes and understanding the logic of their experience brings you fresh insights. Afterwards, if you still enlist disciplinary terms as codes, you will use them more consciously rather than automatically. Thus, you can elect to use only those terms that fit your data.

Advantages of Initial Coding

From the start, careful word-by-word, line-by-line, or incident-with-incident coding moves you toward fulfilling two criteria for completing a grounded theory analysis: fit and relevance. Your study fits the empirical world when you have constructed codes and developed them into categories that crystallize participants' experience. It has relevance when you offer an incisive analytic framework that interprets what is happening and makes relationships between implicit processes and structures visible.

Careful coding also helps you to refrain from imputing your motives, fears, or unresolved personal issues to your respondents and to your collected data. Some years ago, a young man in my undergraduate seminar conducted research on adaptation to disability. He had become paraplegic himself when he was hit by a car while bicycling. Stories of courage, hope, and innovation filled his ten in-depth interviews. Narratives of grief, anger, and loss permeated his analysis of them. After I noted that his analysis did not reflect his collected material, he realized how his feelings had colored his perceptions of other people's disabilities. His was an important realization. However, he might have arrived at it before he handed in his paper had he done more assiduous coding. Line-by-line coding might have changed his ideas about his data early in the analysis.

Coding forces you to think about the material in new ways that may differ from your research participants' interpretations.

Your analytic eye and disciplinary background lead you to look at their statements and actions in ways that may not have occurred to them. You may forge links between their actions and larger social processes that take your emerging grounded theory beyond the immediate individuals or observational setting. Salespeople in a US department store, for example, may be concerned with struggling to meet their managers' sales quotas and keeping their jobs. They may not think about how such everyday work policies and practices serve globalizing processes.

By studying the data and following leads you find in them, you may make fundamental processes explicit, render hidden assumptions visible, and give participants new insights. Thomas (1993) says that a researcher must take the familiar, routine, and mundane and make it unfamiliar and new. Think of seeing a once familiar landscape with a fresh eye after a long absence. You see familiar landmarks with acuity, unlike days past when they blurred together. Word-by-word and line-by-line coding help you to see the familiar in new light. Incident-with-incident coding aids you in discovering patterns and contrasts. You may gain surprising insights about how people's actions fit together or come into conflict. You also gain distance from your preconceptions and your participants' taken-for-granted assumptions about the material so that you *can* see it in new light.

In Vivo Codes

Grounded theorists refer to codes of participants' special terms as *in vivo* codes. These terms provide a useful analytic point of departure. *In vivo* codes help us to preserve participants' meanings of their views and actions in the coding itself. *Pay attention to language while you are coding. In vivo* codes serve as symbolic markers of participants' speech and meanings. Whether or not they provide useful codes in the later more integrated analysis depends on how you treat them analytically. Like any other code, they need to be subjected to comparative and analytic treatment. Although participants' terms may be catchy, *in vivo* codes do not stand on their own in a robust grounded theory; these codes need to be integrated into the theory. When you scrutinize them carefully, four kinds of *in vivo* codes prove to be useful:

- **Terms everyone 'knows' that flag condensed but significant meanings**
- **A participant's innovative term that captures meanings or experience**
- **Insider shorthand terms reflecting a particular group's perspective**
- **Statements that crystallize participants' actions or concerns.**

In vivo codes use research participants' terms as codes to uncover their meanings and understand their emergent actions. Linda Ziegahn and Kathleen A. Hinchman's (1999) in vivo codes – 'breaking the ice,' 'figuring out how to help,' 'trying to understand' – provided form and coherence to their study of college students who tutored adult learners. They use gerunds to code and thus portray problems that the tutors wrestled with. Despite the student tutors' increased awareness of poverty and lack of opportunity, the authors learned 'that the border between campus life and the adult literacy community is a site of reproduction rather than transformation' (p. 99).

Authors may use *in vivo* codes to title their works. Doig, McLennan, and Urichuk's (2009) title reads '"Jumping through Hoops": Parents' Experiences with Seeking Respite Care for Children with Special needs;' Teti, Bowleg, and Lloyd's (2010) title is '"Pain on Top of Pain, Hurtness on Top of Hurtness": Social Discrimination, Psychological Well-Being, and Sexual Risk among Women Living with HIV/AIDS;' and Alexandra Solomon et al. (2011) adopt the title '"Don't Lock Me Out": Life-Story Interviews of Family Business Owners Facing Succession.' Noel Richardson (2010) also uses an *in vivo* code for his title, '"The 'Buck' Stops with Me" – Reconciling Men's Lay Conceptualisations of Responsibility for Health with Men's Health Policy' [in Ireland]. Richardson picks up this thread and weaves his major *in vivo* code into sections of the article: '"The 'buck' stops with me" – Being a Responsible Citizen' (p. 423); '"It's an eye-opener really" – Responsibility and Fatherhood' (p. 424); '"Things began to creak a little bit" – Ageing and Increased Responsibility for Health' (p. 424); and 'Wake-up call' (p. 425). Richardson's *in vivo* title and sections headings link Irish men's experience and views with tensions and contradictions in Ireland's published statement of a national health policy for men.

When participants assume everyone shares the condensed meanings of *in vivo* codes, they give you grist for analysis. Look for participants' implicit meanings and attend to how they construct and act upon these meanings. In doing so, we can ask, what analytic category or categories does this code suggest? Unpacking such terms gives you an opportunity not only to understand implicit meanings and actions but also to make comparisons between data and with your emerging categories.

Some *in vivo* codes simultaneously reflect condensed meanings of a general term and reveal an individual's fresh perspective. After suffering a sudden onset of a serious chronic condition, one man said he intended to pursue 'making a comeback' (Charmaz, 1973). By borrowing a term from once successful celebrities, he defined his stance toward dealing with chronic illness. Other participants' actions and statements indicated that they shared this stance, although they did not invoke this vivid term.

In vivo codes are characteristic of social worlds and organizational settings. Organizations use a rich array of terms researchers can treat as *in vivo* codes. Calvin Morrill's (1995, pp. 263–268) glossary of executives' terms in one corporation included both general terms and specific labels that no doubt furthered his understanding of how they dealt with conflict. Executives imbued some terms, such as 'bozo,' 'roadblock,' or 'jumping ship,' with meanings that echoed ordinary parlance, although many terms assumed specific meanings within the organization and evoked metaphors of combat, violence, and violation. Morrill's inventory of specialized terms includes the following:

BLACK KNIGHT

An executive who often engages in covert action against opponents, does not support his intra-departmental colleagues in disputes ... ; (in take-over imagery, black knight

refers to an unfriendly acquirer from the perspective of an acquired firm). (p. 263)

FLYING LOW
Not confronting an offender with longstanding grievances against their behavior. (p. 265)

VAPORIZING
Terminating an executive from the company or creating the conditions under which an executive resigns from the corporation. (p. 267)

At organizational or collective levels of analysis, *in vivo* codes reflect assumptions, actions, and imperatives that frame action. Studying these codes and exploring leads in them allows you to develop a deeper understanding of what is happening and what it means. Such codes anchor your analysis in your research participants' worlds. They offer clues about the relative congruence between your interpretation of participants' meanings and actions and their overt statements and actions. *In vivo* codes can provide a crucial check on whether you have grasped what is significant.

In each study you conduct, participants will word or write things in ways that crystallize and condense meanings. Hearing and seeing their words allows you to explore their meanings and to understand their actions through coding and subsequent data collection. Pursue telling terms. One young doctor with severe diabetes explained himself as being 'supernormal' (Charmaz, 1973, 1987). As our conversation unfolded, his meaning of supernormal became clear. Not only did he intend to manage being a physician without his condition deterring him, he also aimed to excel beyond his peers. His hopes and plans symbolized identity goals in social life that transcended psychological predilections. Once I grasped the idea of pursuing supernormal identity goals, I saw this process reflected in other participants' actions and stated intentions. Similarly, other *in vivo* codes emerged as I heard many people advocate 'taking one day at a time,' and listened to their stories of having 'good days' and 'bad days.' Subsequently, I sought the condensed meanings and actions that these terms covered and coded for them.

Transforming Data into Codes

Coding relies on having solid data. How and what you record affects what you have to code. Increasingly, qualitative research draws on in-depth and focus group interviews. A few grounded theorists advocate coding from notes rather than transcribed interviews. Presumably, you grasp the important points and eliminate clutter. This approach assumes an objective transparency of what participants say and do. It also assumes that any keen interviewer will recall and record the most telling material and record it well, an unsupported assumption that Gary Fine's (1993) experiment refutes. He found that even trained ethnographers rapidly forgot details. Interviewers who rely solely on notes may further assume that their notes and codes have 'captured' the participants' views and actions. None of these assumptions may be true – even for experienced researchers.

Coding full interview transcriptions gives you ideas and understandings that you otherwise miss. Thus, the method of data collection not only forms your materials, but also frames your codes. Coding full transcriptions can bring you to a deeper level of understanding. In contrast, coding from and across notes might give you a wider view. It also can, however, contribute to grounded theorists going around the studied phenomenon rather than into it. An overriding emphasis on plausibility that sidesteps thoroughness and systematic study risks constructing superficial analyses.

Transcribing entire interviews and fieldnotes also has some hidden benefits. Your first reading and coding of the data need not be the final one. Rich, thorough data can generate many research questions. Such data contain the makings of several analyses, whether or not you realize it early in your research. You can save a set of related codes to develop later. You can return and recode a set of old data. In both cases, your codes spark new ideas. In the meantime, the full recordings preserve details for these ideas to ignite later. You may be amazed at the diverse ideas you can gain from the data for one project.

Any method of data collection frames what you can code. Ethnographers sometimes rely more on what they hear than what they see, and interviewers may rely on only what they hear. Record what you see as well as what you hear. An interviewer sees a scene and at least one person. Notes about these observations are data to code. In one of Abdi Kusow's (2003) interviews of Somali immigrants, his observations constituted most of his data. Kusow had already found that many potential participants declined to be interviewed because of the volatile political climate in Somalia. One participant referred him to a young woman who consented to the interview. When Kusow arrived, the TV was blaring and she and several small children were watching it. She did not suggest leaving the room, kept the TV on, and answered his questions in monosyllables. Kusow saw her responses as 'basically her way of giving me no information at all' (p. 596). Kusow's anecdote suggests a dictum for interviewers: Code your observations of the setting, scene, and participant as well as your interviews. Revealing data resides in such observations.

Concluding Thoughts

Initial coding routes your work in an analytic direction while you are in the early stages of research. You can make grounded theory coding familiar through practice, and then evaluate how it works for you. By remaining open to the data as you had been open to statements, observations, and events in the research setting, you will discover subtle meanings and have new insights. I recommend completing a close initial coding at the level that best fits your data and task.

Grounded theory coding is part work but it is also part play. We play with the ideas we gain from the data. We become involved with our data and learn from them. Coding guides us in viewing data. Through coding we make discoveries and gain a deeper understanding of the empirical world.

Theoretical playfulness allows us to try out ideas and to see where they may lead. Initial coding gives us direction and a preliminary set of ideas that we can explore and examine analytically by writing about them. Grounded theory coding is flexible; if we wish, we can return to the data and make a fresh coding. We can go forward to writing about our codes and weighing their significance.

Initial coding is that first part of the adventure that enables you to make the leap from concrete events and descriptions of them to theoretical insight and theoretical possibilities. Grounded theory coding surpasses sifting, sorting, and synthesizing data, as is the usual purpose of qualitative coding. Instead grounded theory coding begins to unify ideas analytically because you kept in mind what the possible theoretical meanings of your data and codes might be. Now that you have some codes, it is time to proceed to studying, sorting, and selecting them. The next chapter outlines ways to engage in focused coding and advanced forms of coding.

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1 The data about Teresa's experience of having anaplastic cancer forms the basis of a methodological demonstration project, *Five Ways of Doing Qualitative Analysis: Phenomenological Psychology, Grounded Theory, Discourse Analysis, Narrative Research, and Intuitive Inquiry* (Wertz, Charmaz, McMullen, Josselson, Anderson, and McSpadden, 2011) in which each qualitative methodologist used his or her method to analyze the same data. This data consisted of a personal account of an unfortunate event and a subsequent interview about it from one main participant, "Teresa," and analogous materials from another participant, "Gail," for those of us who requested more data for comparative purposes. I also made comparisons with interview data from my earlier studies. Even though we had limited materials to demonstrate our diverse analytic approaches, each methodologist developed a cogent analysis that exemplified his or her analytic method. We only aimed to demonstrate how we used our respective methods but did discuss what we would ask, how many participants we might seek, and which directions we would take to complete a full-scale study.

2 From a grounded theory perspective, extant concepts must earn their way into the analysis (Glaser, 1978; Glaser & Strauss, 1967).

3 For clarification, the full names are Karl Marx, George Herbert Mead, Michel Foucault, and Sandra Harding.

4 Discovering that the data have holes is not limited to qualitative research. Survey researchers who conduct standardized interviews sometimes discover when conversing with respondents afterwards that their questions do not tap significant areas. Quantitative researchers must stick to the same instrument but qualitative researchers can remedy such problems while gathering data. So what might quantitative researchers do? Conducting qualitative research on a topic before constructing quantitative instruments can help. The current turn to mixed methods will likely bring more grounded theorists into quantitatively-driven projects.

5 By 1992, Glaser seems to disavow line-by-line coding as he advises against taking apart a single incident. He states that line-by-line coding produces a `helter skelter' of over-conceptualizing the incident and generates too many categories and properties (p. 40) without yielding an analysis. Nonetheless, a researcher can select the most telling codes

gained through line-by-line coding of an incident and make comparisons between incidents.

6 Students often think observing behavior in public places is the easiest type of qualitative research to conduct. Not so. Both the researcher's data and analytic approach make a difference. Few novices have the eye and ear to record nuances of action and interaction that constitute the tiny incidents shaping routine events. More likely, they record concrete behaviours in a general way and gradually learn to make more acute observations.